http://bit.ly/fileOrgSlides

Data Management: File Organization

2024-01-22 | MIT Libraries Data Management Services

data-management@mit.edu

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Introductions

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Participation

We'll be going through various exercises together and posting ideas in chat.

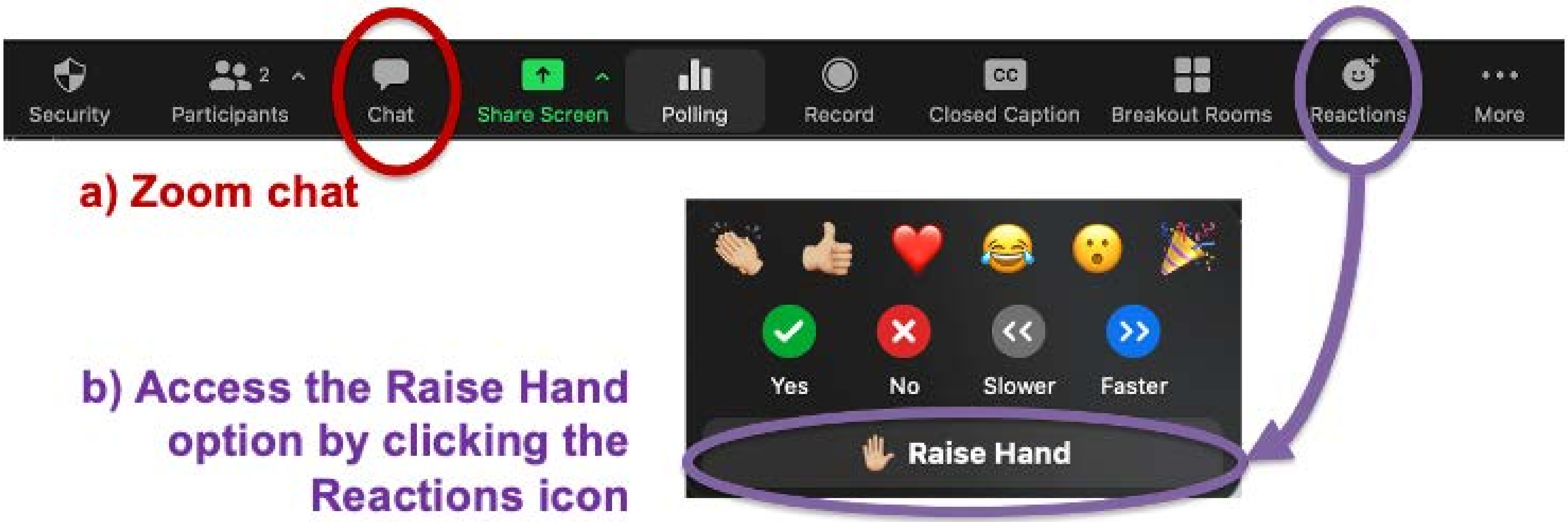
If you don't want to share your file organization process with the group or you want to go more in depth later, you can access a more detailed word doc file organization worksheet at http://bit.ly/fileOrgWorksheet

Ask questions

Ask questions by either

a) typing them in the **Zoom chat window**, or

b) use the **Raise Hand** option under the **Reactions icon** at the bottom of your Zoom window.



Data Management Services @ MIT Libraries

Workshops

- Upcoming Classes & Events: https://libraries.mit.edu/news/events/

Web guide

- http://libraries.mit.edu/data-management

Consultations (individual or research groups)

- includes help with creating data management plans, setting up file organization, finding data repositories, etc.

Contact: data-management@mit.edu

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Today's session

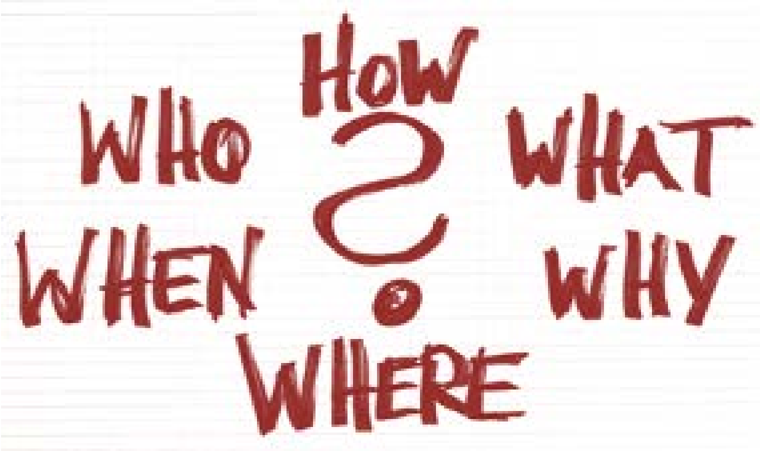
Why & When & Who?

What?

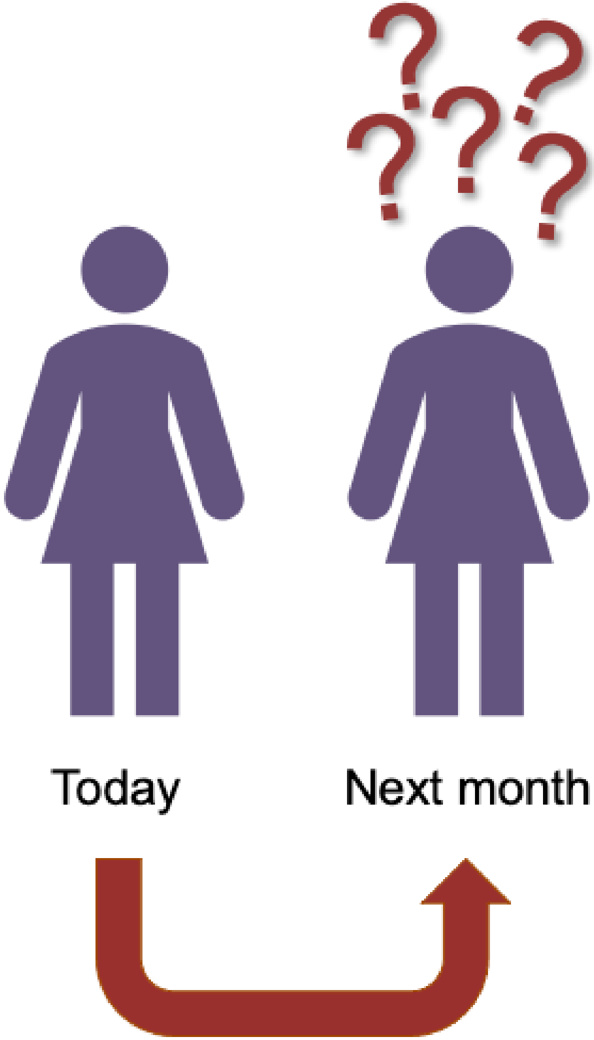
How?

- Naming conventions
- Versioning
- Organizational structures

Who & Where?



Why file organization is important



The first person with whom you will share your data is yourself.

Why file organization is important

Work is messy.

In time you will create
multiple files in
various formats,
multiple versions,
methodologies, etc.,
all relating to your research.

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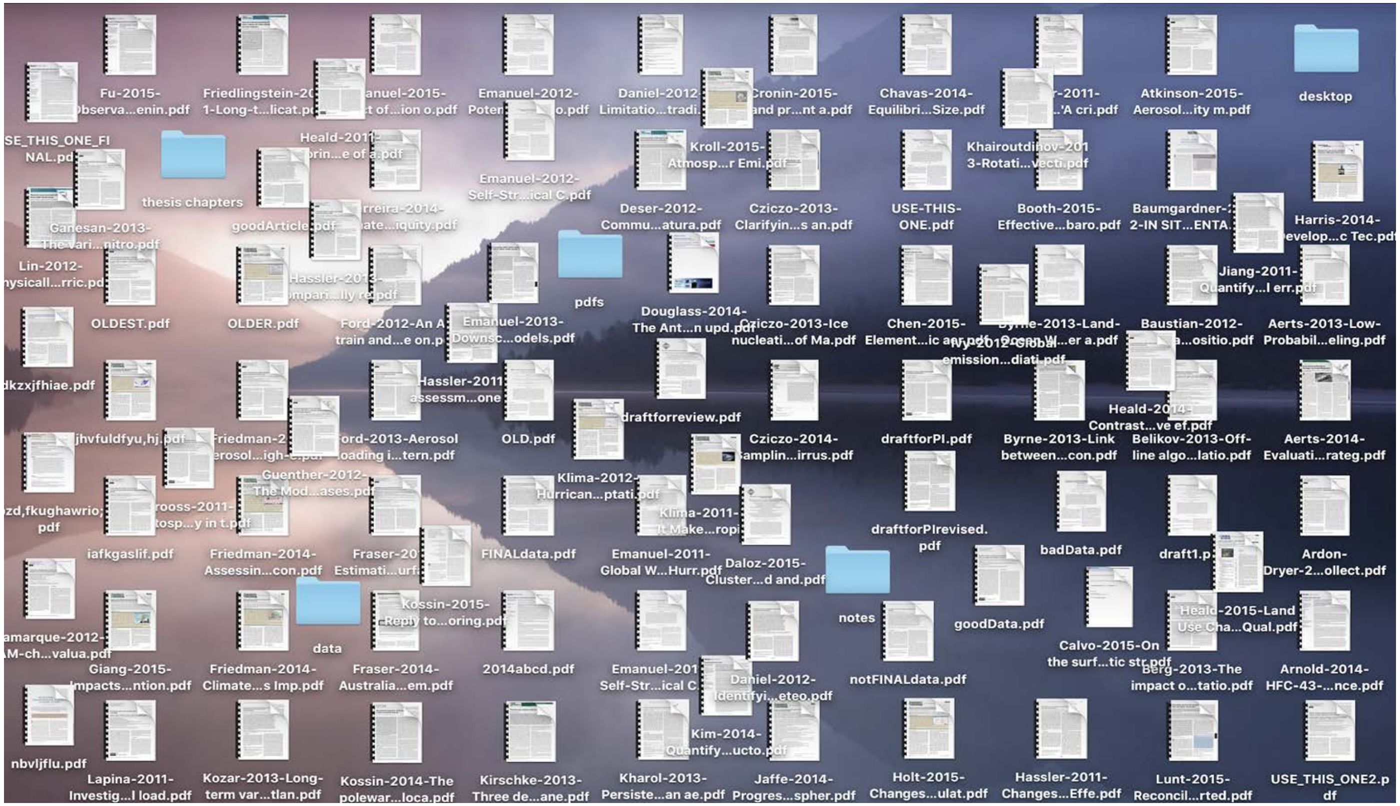
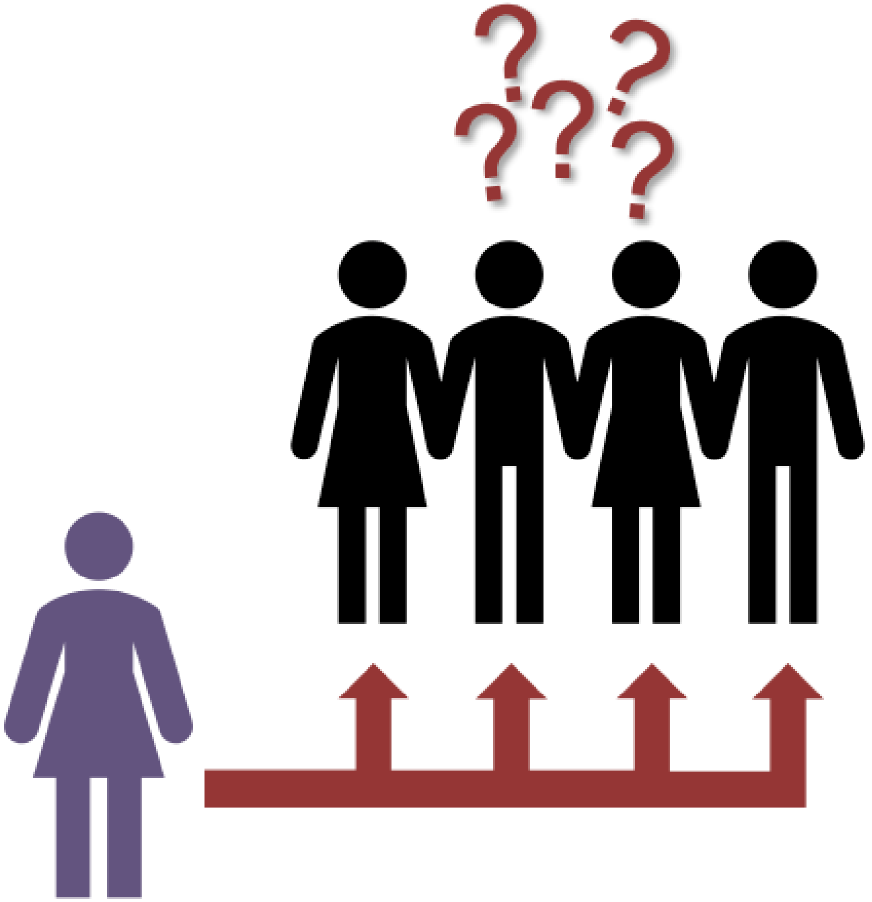
Why file organization is important



Why file organization is important

Can someone else
(& their machines)
understand & use your files?

Now?
Tomorrow?
In 5 years?



Key principles of file organization

- Spending a little **time upfront** can save a lot of time later on.
- Be realistic: **strike a balance** between doing too much and too little.
- There's **no single right way** to do it; establish a system that works for you.
- Think about **who your system needs to work for**: Just you? You and your lab group? Collaborators?

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Your turn!

Part A. Defining your organizational needs

- How do you currently organize your files?
- Reflect on this current system.
  - What works and what doesn't work?
  - What do you like / don't like about your system?
- Who are you designing your organization system for?
  - List the people that need to access the files in your system.
  - Any organization preferences or needs of this group?

Today's session

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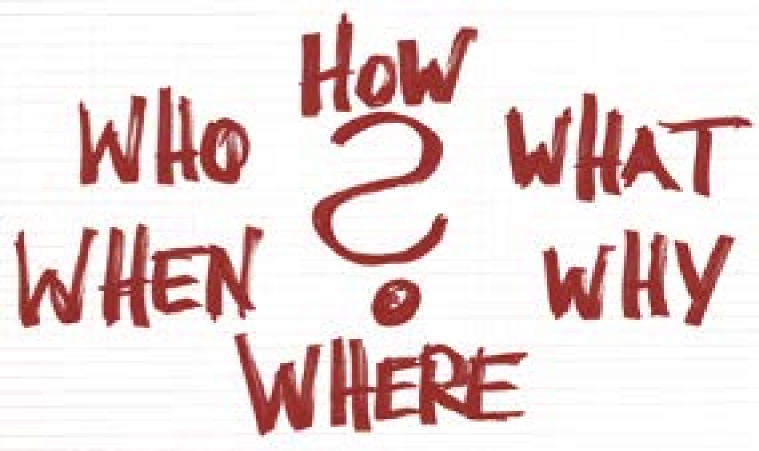
Who & Where?



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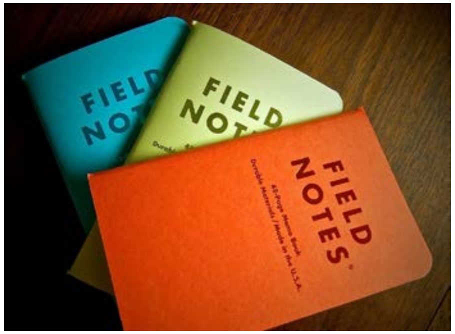
What are you organizing?

- Instrument & experimental data
- Progress reports & presentations
- Field observations
- Secondary data
- Analysis files & graphics
- Microscopy images
- Literature
- …



Credit: Katie McNeil, former DMS member

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What are you organizing?

Start small!

- 1 project
- 1 grant
- …

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Your turn!

Part B. Creating a file naming schema

Create your file inventory.

- Start small. Consider just a project or just an experiment and then expand from there.
- What types of data are you working with and what file formats are they in?

Today's session

Why & When & Who?

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How?

- **Naming conventions**
- Versioning
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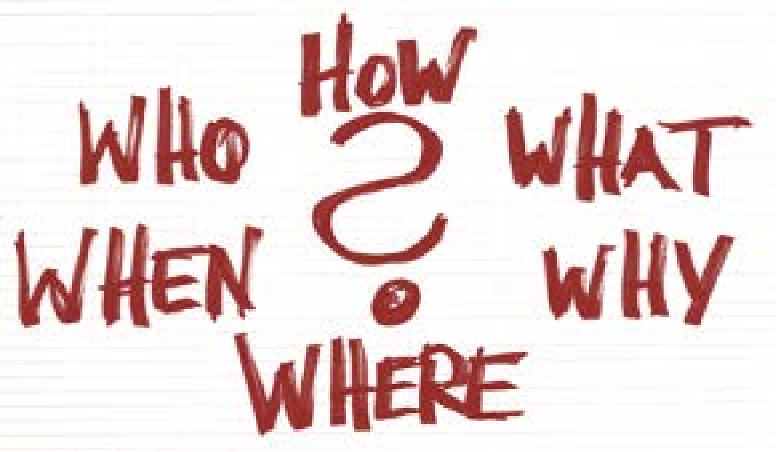
Who & Where?



Naming: be descriptive

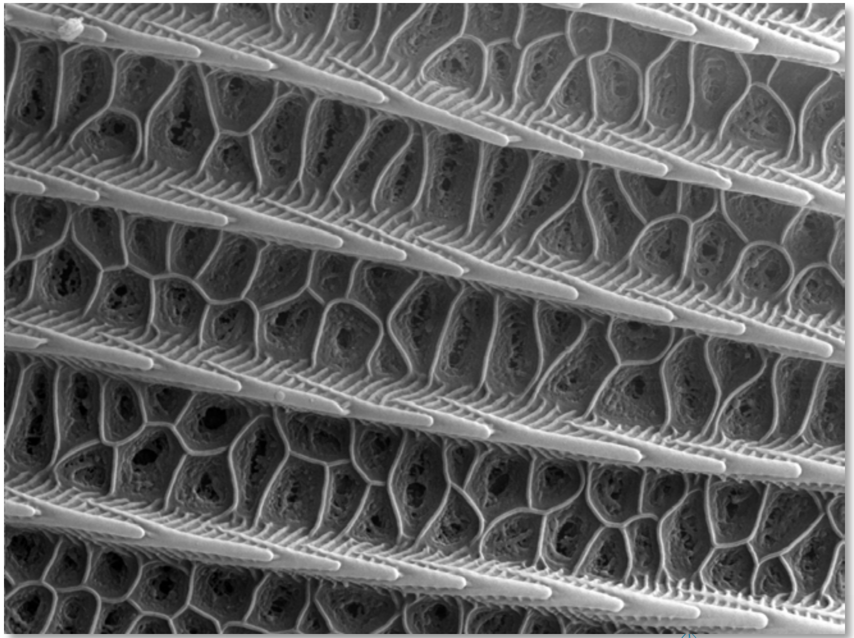
What do you include?

[?]_[?]_[?].ext

- What elements ***describe the file***?
  What keywords would you use to search for it?
  - Unique identifier (e.g., grant number, sample ID)
  - Project or research data name
  - Conditions (lab instrument, solvent, temperature, etc.)
- In a set of similar files, what ***makes each file unique***?
  - Run of experiment (sequential number)
  - Date
  - Version number

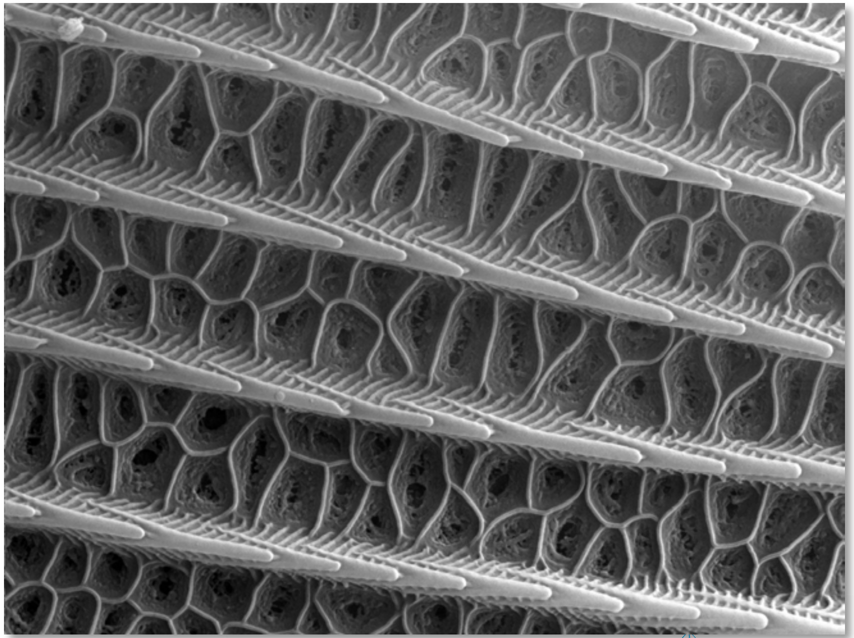
File naming example, start with…

abcdefghijklmnopqrstuvwxyz.sam



File naming example, change to…

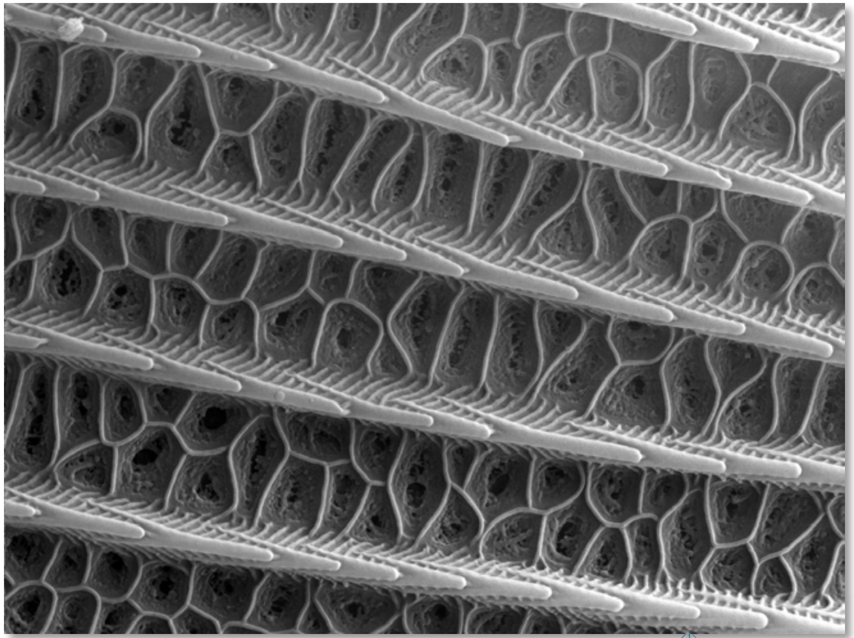
sam_monarch_wing_20170115_CM_001.sam
sam_monarch_wing_20170115_CM_001.tif



File naming example, change to…

sam_monarch_wing_20170115_CM_001.tif

[instrument]_[item]_[part]_[date]_[collector]_[ascension#].ext



Naming: be consistent

Keep the same information in the same order

✓ [instrument]_[item]_[part]_[date]_[collector]_[ascension#].ext
✓ **sam_monarch_wing_20170115_CM_001.tif**
✓ **sam_monarch_wing_20170115_CM_002.tif**
✓ **sam_monarch_wing_20170115_CM_157.tif**

✗ **sam_monarch_wing_20170115_CM_001.tif**
✗ **monarch_wing_ sam_ 20170115_CM_002.tif**
✗ **monarch_wing_20170115_ sam_ CM_157.tif**

Naming: be consistent

Do this	Not this
***Same* information in the same order**	ProjectID_Collector Date_ProjectID Time_ProjectID
***Same* date format**	YYYYMMDD MMDDYYYY YYMMDD
***Same* abbreviations and acronyms**	sam_monarch_wing samicro_monarch_wing sam_m_wing
***Same* number of leading zeroes for sequential numbers**	Sample_001234 Sample_01234 Sample_1234

File naming examples

[class]_[material]_[date].ext
DataViz101_Slides_20210427.pptx
DataViz101_Slides_20210720.pptx
DataViz101_toChange_20210720.txt

[project]_[state]_[year]_[dataset]_[analysisID].ext
SevilletaLTER_AZ_2001_NPP_201707.csv
SevilletaLTER_NM_2001_NPP_201707.csv
SevilletaLTER_NM_2002_NPP_201707.csv

File naming best practices

1. Don’t use spaces.

- Machines can get confused by spaces.
- Harder to write code when files have spaces.

2. Don’t use special characters.

- Some characters already carry meanings (^ . ? + | $).
- It’s difficult to visually distinguish between hyphens, en dashes, & em dashes, but machines treat them differently.

✗ Progress report: CM (expt 0159).pptx

✗ Progress report “expt 0159” 04-27-2011.pptx

Only use numbers, letters, underscores.

✓ expt0159_progReport_20210427.pptx

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File naming best practices

3. Don't depend on letter case alone to distinguish between files.

- Some systems treat ***a*** and ***A*** differently. Some don't.
- camelCase can make it easier to read file names, but always include a unique non-case dependent element.

✗ progReport_CM.pptx
✗ progReport_cm.pptx

✗ progReport_20210427.pptx
✗ progreport_20210427.pptx

✓ progReport_CM_20210427_v1.pptx
✓ progReport_CM_20210427_v2.pptx

File naming best practices

4. When using sequential numbering, use leading zeros.

- For a sequence of 1-10: 01-10
  For a sequence of 1-100: 001-010-100
- This allows files to sort well & have consistent name lengths.

✗ sam_monarch_wing_20170115_CM_**1**.tif
✗ sam_monarch_wing_20170115_CM_**2**.tif
✗ sam_monarch_wing_20170115_CM_**157**.tif

✓ sam_monarch_wing_20170115_CM_**001**.tif
✓ sam_monarch_wing_20170115_CM_**002**.tif
✓ sam_monarch_wing_20170115_CM_**157**.tif

File naming best practices

5. Only use a period before the file extension.

- Extra periods can confuse both machine and human readers.

  ✗ name.date.ext
  ✗ name.v1.1.ext
  ✓ name_date.ext
  ✓ name_v1a.ext

6. Limit file names to 32 characters or less.

- The longer the filename, the longer the file path to type.
- Shorter names are easier to quickly scan to know what file is.

  ✗ thisIsTheFileNameThatDoesntEndItJustGoesOnAndOnMyFriend.ext
  ✓ 32CharactersLooksExactlyLikeThis.ext
  ✓ shorter_is_better.ext

File naming best practices

See our handout version.

Be descriptive. (“final” is NOT descriptive)

Be consistent.

1. Don’t use spaces.
2. Don’t use special characters.
3. Don’t depend on letter case alone to distinguish between files.
4. When using sequential numbering, use leading zeros.
5. Only use a period before the file extension.
6. Limit file names to 32 characters or less.

File naming & instruments

Check if your instrument, software, or other equipment that outputs your data files can be set with a file naming system.

This is less work than retrospectively changing filenames.

But if you still have to change many file names downstream…

Credit: Katie McNeil, former DMS member

File naming & batch/bulk renaming

You can use tools that retrospectively align file/folder names with naming conventions.

Caveats:

- Ideally you want to be able to map the original to new names
- Make sure it doesn't change the file extension

Batch File Renaming Tools:

- See **our handout** for some common tools

Your turn!

Part B. Creating a file naming schema

Decide what to include in your filenames.

- What are the unique characteristics you need to put in the file name
  - to easily recognize what the file is?
  - to find the file you’re looking for in the future?
  - to distinguish your files from each other?

Today's session

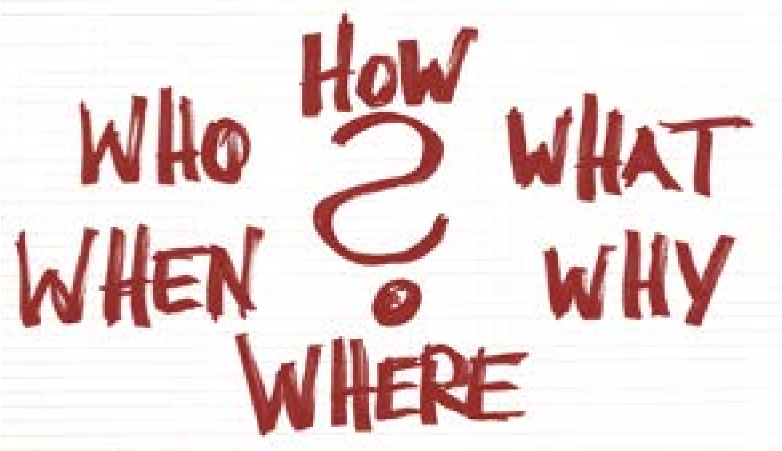
Why & When & Who?

What?

How?

- Naming conventions
- **Versioning**
- Organizational structures

Who & Where?



Versioning: *why?*

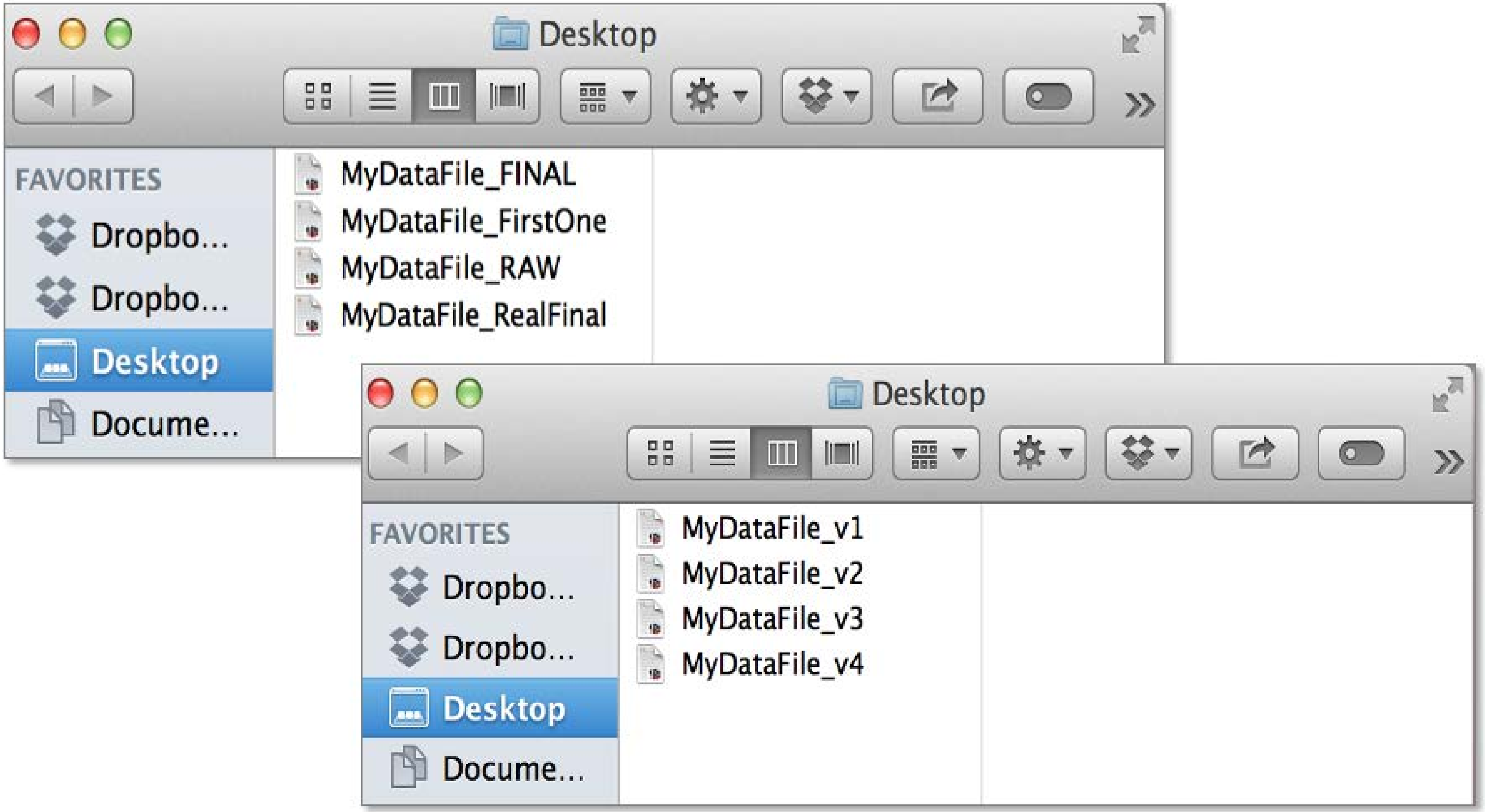
Versioning helps you to easily find the version you want and not overwrite the version you need.

https://twitter.com/GrammarGirl/status/1443640746905505813

http://bit.ly/fileOrgSlides

Versioning: *why?*

Versioning helps you to easily find the version you want and not overwrite the version you need.



Versioning: *when?*

Depending upon practices in your field, version either:

- Analysis/program/script files
- Data files themselves

Don’t forget to also version:

- Project documentation
- Progress report drafts
- Etc.

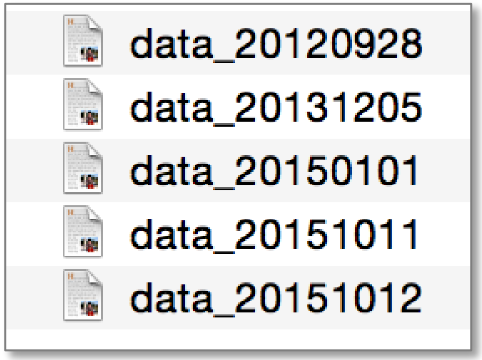
Versioning: *how?*

1. Establish a consistent convention
2. Document your convention
3. Share your convention

See our version control handout
for more information & resources!

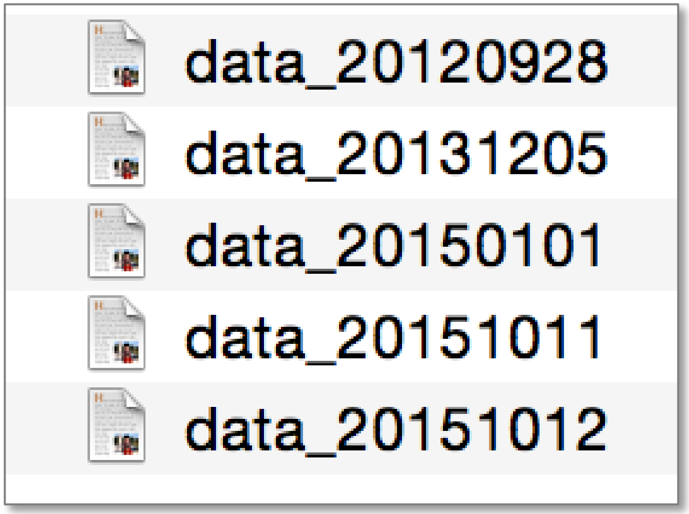
Versioning: *levels of complexity*

Basic – file names



Versioning: *using dates*

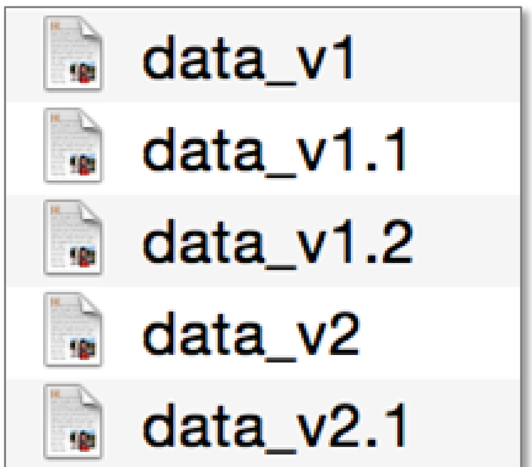
Use dates to distinguish between successive versions.

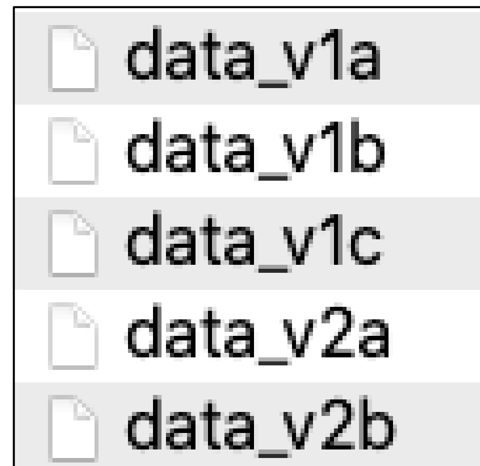


This is not ideal when you can potentially have multiple versions in a day.

Versioning: *using version numbers*

Use ordinal numbers (1, 2, 3, etc.) for major version changes and **a letter*** for minor changes.





* A decimal number was previously used for minor changes, but the decimal breaks good file naming conventions.
Better practice is to eliminate minor change distinctions altogether.

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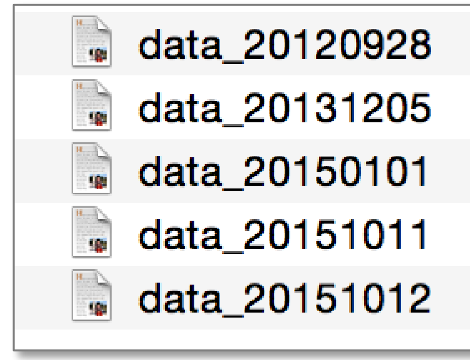
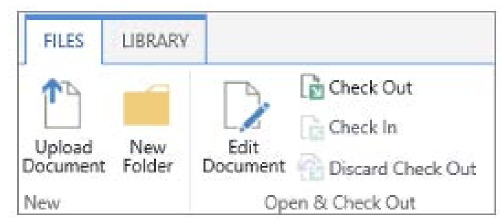
Versioning: *levels of complexity*

Basic – file names



Intermediate – built-in software capabilities



More advanced – version control software

Versioning: *documentation*

In many cases, it is helpful to log the changes so that you can quickly assess and access the versions.

It's good to document:

- What was changed?
- Who is responsible?
- When did it happen?
- Why?

CHANGELOG.txt

git

CHANGELOG.md

Versioning: *documentation*

Basic Changelog Template

This template shows the basic elements of a changelog and one way of simply documenting changes in file versions.

fileName_Changelog

All notable changes to this project/file will be documented in this log.

v1 YYYY-MM-DD J Doe <jdoe@example.com>

* change 1 note
* change 2 note

v2 YYYY-MM-DD J Doe <jdoe@example.com>

* change 1 note
* change 2 note

v3 YYYY-MM-DD J Doe <jdoe@example.com>

* change 1 note
* change 2 note

Additional changelog resources:

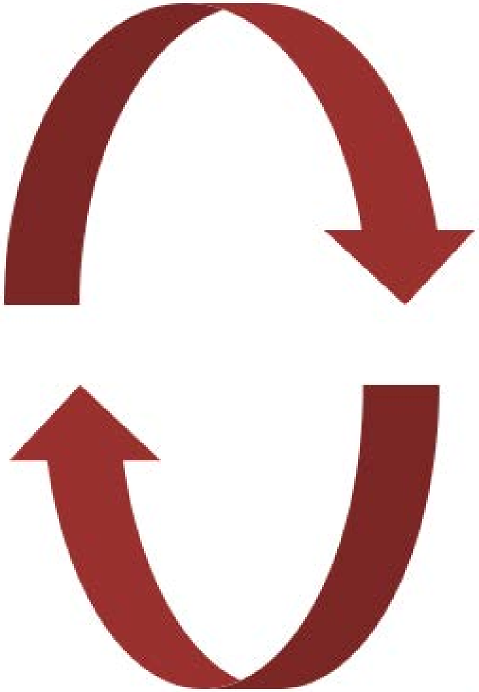
[Keep a Changelog](https://keepachangelog.com/en/1.0.0/)

[Semantic Versioning](https://semver.org/spec/v2.0.0.html)

[Changelog](https://en.wikipedia.org/wiki/Changelog)

Syncing

Be careful when syncing across platforms & simultaneously editing!



Your turn!

Part B. Creating a file naming schema

- Draft your file names
  - Identify **unique characteristics** of each data file type
  - Determine if you need to account for **different versions**
  - Consider what **order** you want the files to sort in
  - What **abbreviations** can you use for your unique elements?

Your turn!

File naming exercise

We present this workshop a few times each year, and we need to save this version of the Power Point slides on the Data Management Services shared drive.

Come up with a filename we could use

- File Organization workshop, one of many workshops we give
- Presented on January 22, 2024
- Presented by DMS team members Sadie Roosa and Amy Nurnberger

http://bit.ly/fileOrgSlides

Today's session

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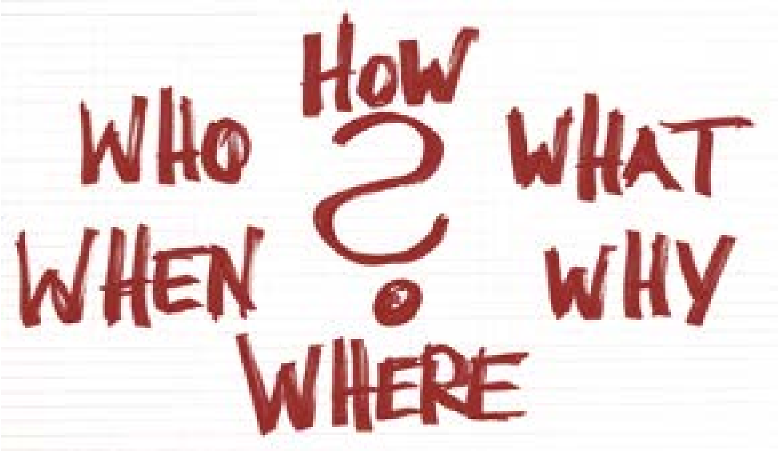
How?

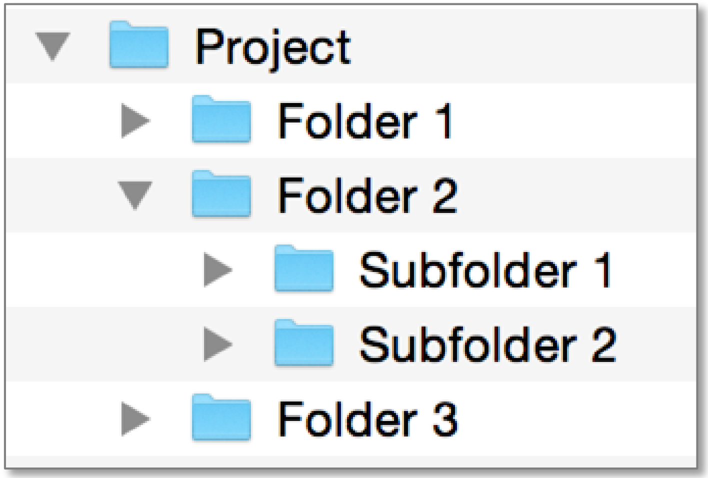
- Naming conventions
- Versioning
- **Organizational structures**

Who & Where?



CC0: https://www.maxpixel.net/Why-Where-Questions-How-Who-What-1328351

Organization structures: *folders & subfolders*



Benefits:

- Familiar & widely used
- Good at representing the structure of information
- Similar items are stored together

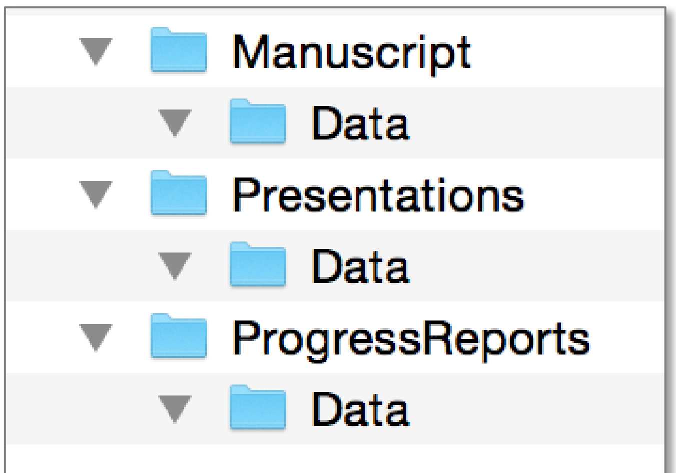
Drawbacks:

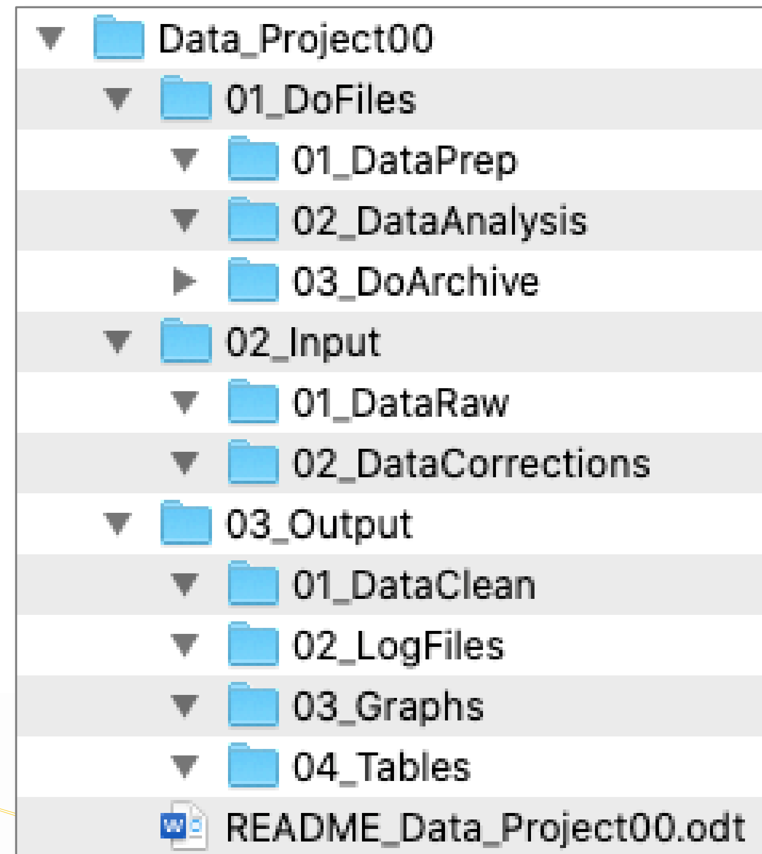
- Surprisingly hard to set up & time consuming to reorganize
- Items can only go in one place
- Challenging to get the right balance between breadth & depth

Organization structures: *folders & subfolders*

Best practices:

- Avoid overlapping categories





Tip: Use numbers as a folder name prefix to order folders how you'd like to view them (e.g., 01_DataClean, 02_LogFiles).

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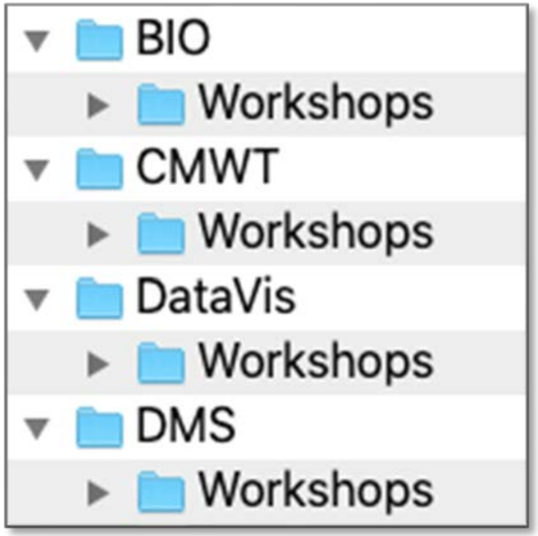
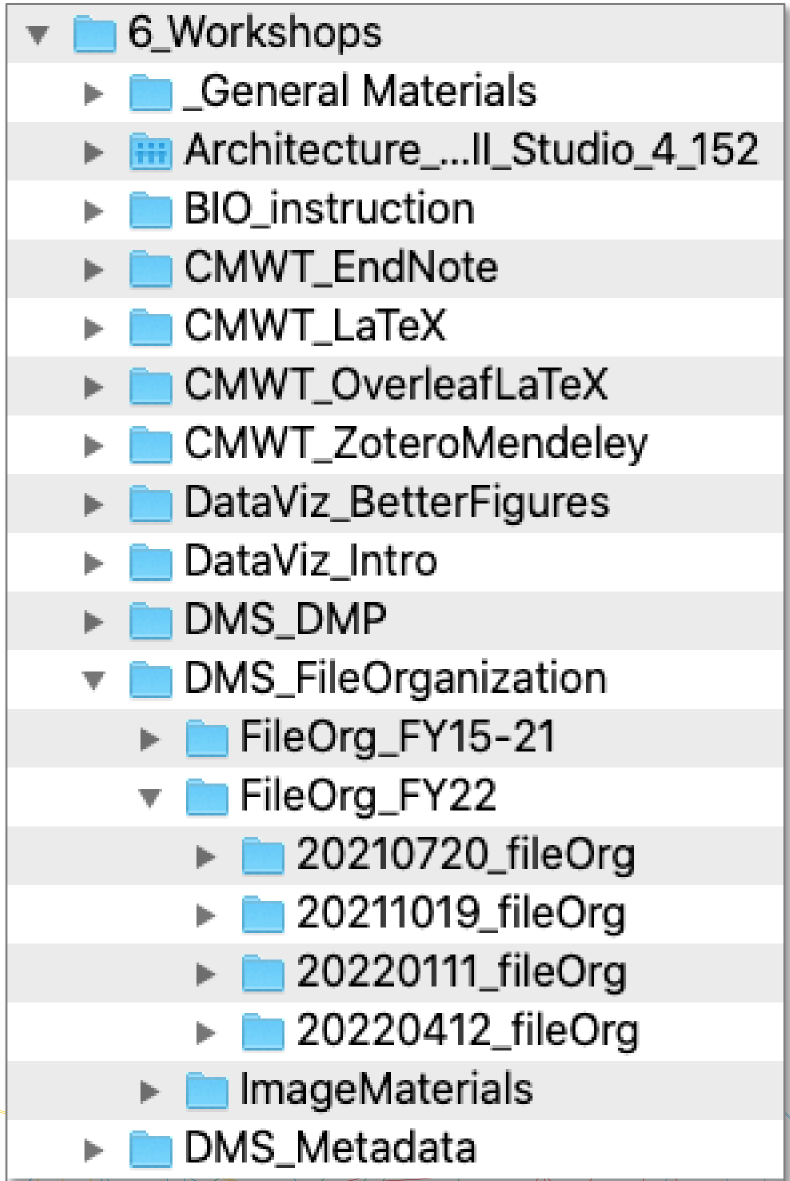
Organization structures: *folders & subfolders*



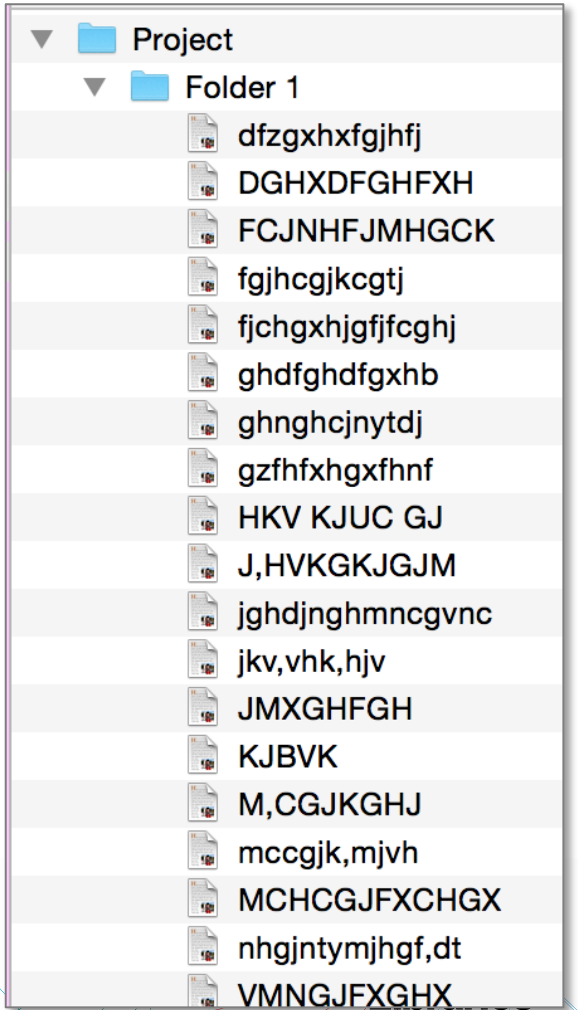


http://bit.ly/fileOrgSlides

Organization structures: *folders & subfolders*

Best practices:

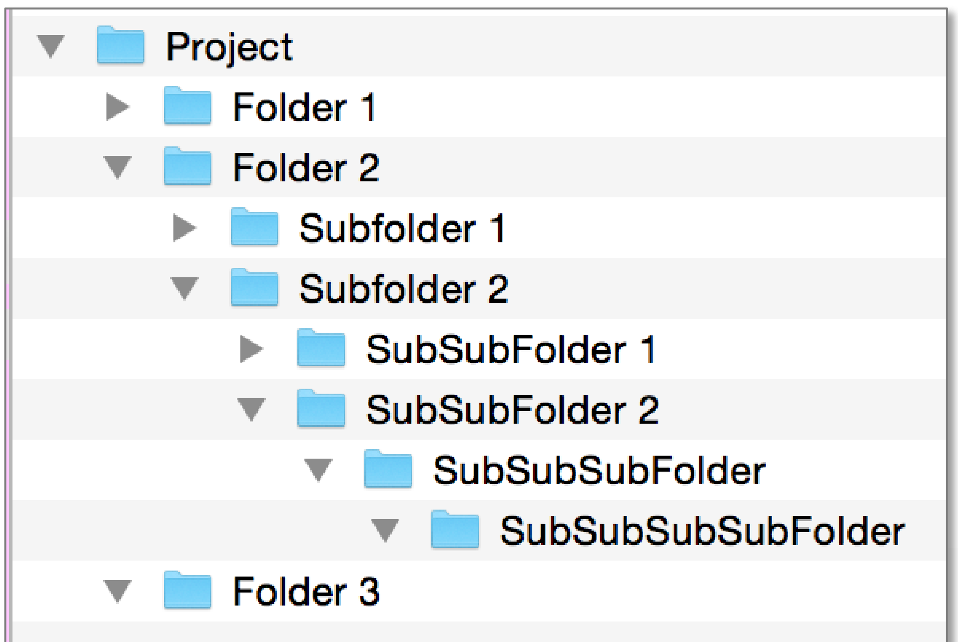
- Avoid overlapping categories
- **Don’t let your folders get too large**



Organization structures: *folders & subfolders*

Best practices:

- Avoid overlapping categories
- Don’t let your folders get too large
- **Don’t let your structure get too deep**



Consider how many clicks it takes to get there or how long a file path may be?

Creating a systematic file folder structure

Steps for defining your system:

✓ Define the types of data and file formats

2. Include important contextual information
3. Organize folders by meaningful categories
4. Choose a directory naming convention

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2. Include important contextual information

When you* are looking for *X file*, how do you think about it?

- By time period?
- By creator/project collaborators?
- By activity or collection method?
- By its type (e.g., presentation, report)?

* Also consider how others in your team think about these files

3. Organize folders by meaningful categories

*There's **no single right way** to do it;*
establish a system that works for you and your files

Primary / Secondary / Tertiary …

[Project] / [Sub-project] / [Experiment] / [Instrument] / [Date]

[Project] / [Project Site] / [Data]

[Project] / [Type of file] / [Data collector name] / [Date]

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3. Organize folders by meaningful categories

[project] / [type of file]

/ butterfly / 1_projectDocs / mtgNotes /

/ butterfly / 1_projectDocs / reports /

/ butterfly / 2_data / 1_input / images / monarch /

/ butterfly / 2_data / 1_input / observations / monarch /

/ butterfly / 2_data / 2_output / monarch /

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4. Choose a directory naming convention

Same approach as file naming.

Determine your unique elements.

Consider ordering and abbreviations.

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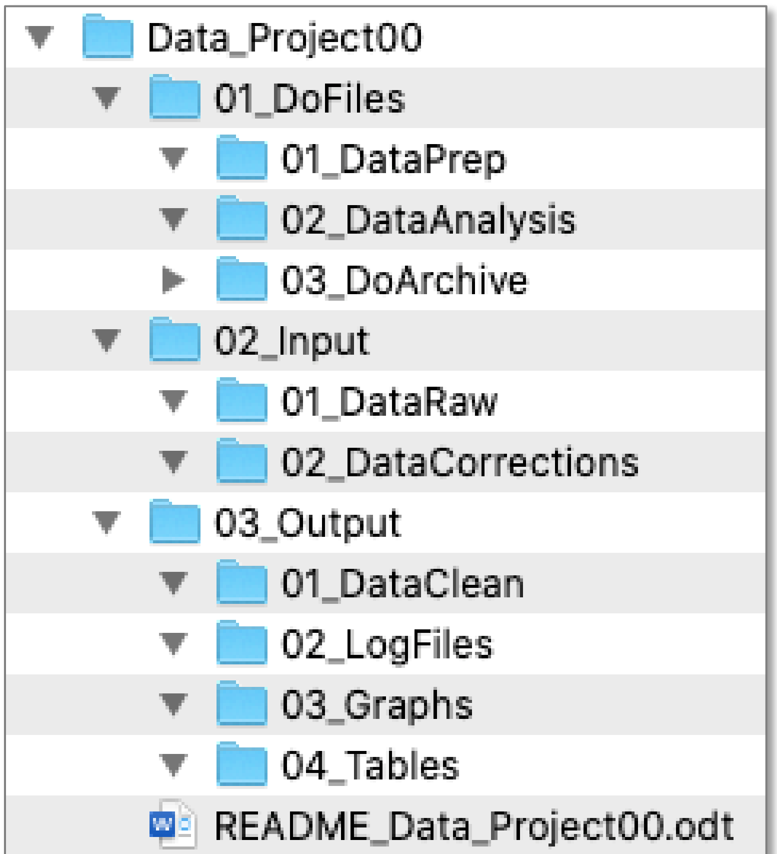
Project structure templates

If you have similarly structured projects, consider creating a ”blank” project template.



- Folder structure & starter files (e.g., templated README) are ready to go
- Copy and tweak to fit project

Example template: https://github.com/djnavarro/newproject/

Your turn!

https://bit.ly/fileOrgDoc

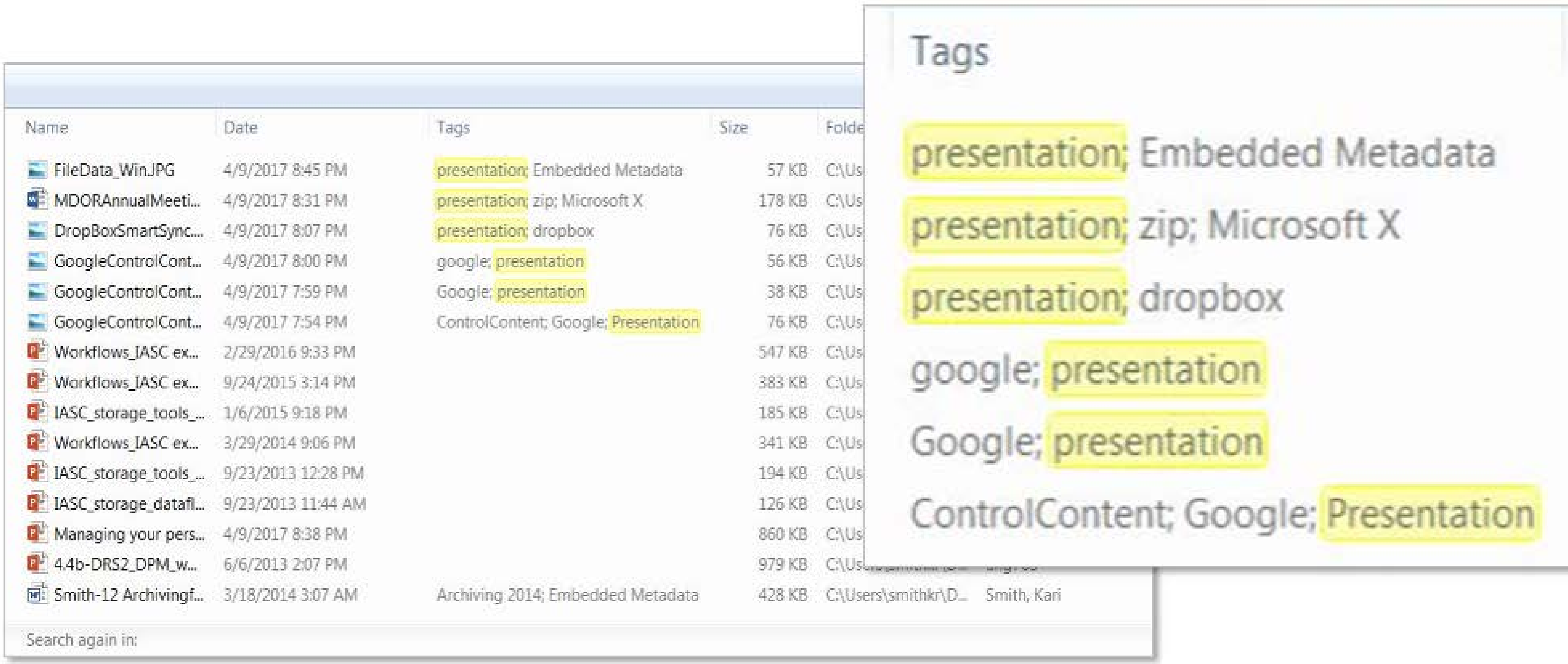
Part C. Creating a systematic file folder hierarchy

- Think about how you would group your files together
  - By file type? By time period? By project? By instrument?
- Use your categories to determine a consistent folder naming convention
- Consider the hierarchy of your folders?
  - How do they relate to each other?

Organization structures: *tags*

Benefits:

- Items can go in more than one category
- Can be quicker/easier to set up
- When collaborating, it can be easier to combine than hierarchical systems



Organization structures: *tags*

Benefits:

- Items can go in more than one category
- Can be quicker/easier to set up
- When collaborating, it can be easier to combine than hierarchical systems

Drawbacks:

- Not how operating systems store files
- If item isn't tagged properly when first acquired, it can be hard to find
- Increased risk of inconsistency
- Less good at representing the structure of information

Organization structures: *tags*

Creating a tag-based system:

1. Determine the **contextual information** by which you want to discover your files
2. Create a **consistent naming convention** for these contextual categories & **document it**!
3. Tag your files!

See our guide to Tagging and Finding Your Files: http://libguides.mit.edu/metadataTools/

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Organization structures: *tag examples*

- Social media platforms (e.g., Twitter, Instagram)
- Journal article keywords
- Notetaking tools (e.g., Evernote)
- Gmail labels
- Citation Management tools (e.g., Zotero, Mendeley)

Speaking of citation management tools…

Citation management tools are software that help you organize your PDFs & citation records.

https://libguides.mit.edu/cite-write

cite-write-tools@mit.edu

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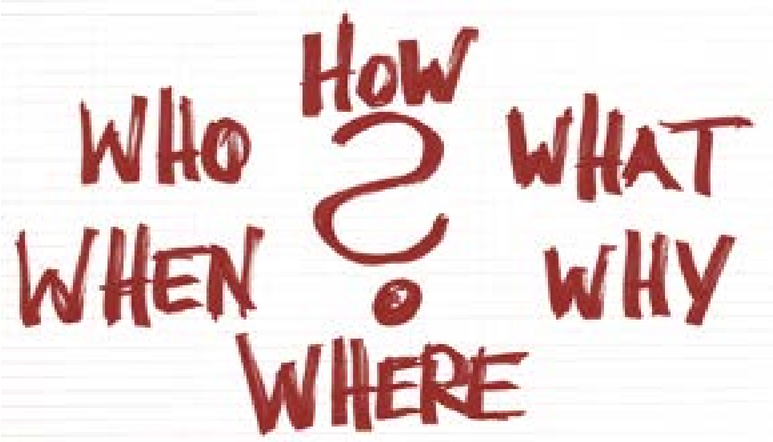
- Organizational structures

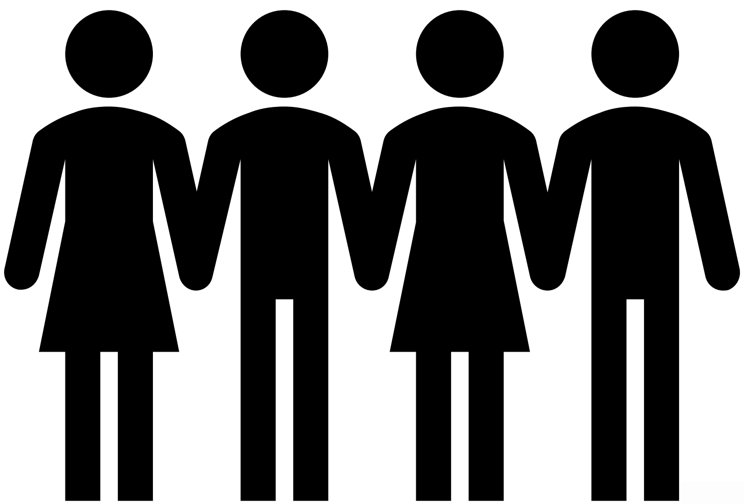
Who & Where?



Who…

…does your system have to work for?

…has responsibility to create / maintain / evolve it?



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README, Workshops_DMS

This README pertains to the files in the shared Workshops_DMS folder.

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GENERAL INFORMATION

Revision history
2018-05-25, Created, Christine Malinowski
...
2021-01-13, Revised, Christine Malinowski (cleaned up formatting)
2021-03-31, Revised, Christine Malinowski (added team GDRive link to Related Documentation)

Files location: Dropbox (MIT)/…/DMS_Team/Workshops_DMS/
README location: Dropbox (MIT)/…/DMS_Team/Workshops_DMS/

Related documentation:
DMS workshops wiki page: https://wikis.mit.edu/confluence/x/haahBw
DMS public website, workshops page: https://libraries.mit.edu/data-management/services/workshops/
DMS_Team shared GDrive, workshops folder: https://drive.google.com/drive/folders/1XQahO9s2AUetKaQ5Mpll7ilLAriq09ec?usp=sharing

FOLDER STRUCTURE

0_public: for materials to be shared externally (e.g., the DMS website Workshops page)
1_generalMaterials: for general materials such as template registration & feedback forms
2_rdm101: for RDM 101 workshops
3_fileOrg: for File Organization workshops
4_versionControl: for Version Control workshops
...

New folders can be added as new classes are developed. The one-offs folder should always be the last folder listed
An OLD folder may be added to each of the folders if it is deemed necessary to separate older versions workshop materials.

FILE NAMES

For general audience workshops:
Filename schema: <topic>_<document type>_<date>.ext
Schema key:
<topic> refers to the workshop, e.g., fileOrg, versionControl, RDM101, in camelCase format *
<document type> refers to the nature of the document, e.g., Handout, Worksheet, Slides
<date> is the workshop date in YYYYMMDD format
Example filename………… FileOrg_Slides_20170718

For one-off or special topics workshops (stored in 99_OneOffs folder), there may be an additional descriptive element to indicate audience:
Filename schema <topic>_<audience>_<document type>_<date>.ext
Schema key
<topic> refers to the workshop, e.g., fileOrg, versionControl, RDM101, in camelCase format *
<audience> refers to the DLC or other audience unit
<document type> refers to the nature of the document, e.g., Handout, Worksheet, Slides
<date> is the workshop date in YYYYMMDD format
Example filename………… FileOrg_Humanists_Slides_20180315.pptx

* For compound file name elements, utilize camelCase when possible or underscore when a visual space break is needed.

http://bit.ly/fileOrgSlides

Documenting your system

Create a README file to document your file organization system.

See our sample and template to get started.

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http://bit.ly/fileOrgSlides

Where…

…do you keep your documentation?

…do you keep your stuff?

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Questions? Comments? Tips?

Check out our web site:
http://libraries.mit.edu/data-management

Contact: data-management@mit.edu

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